

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\ a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a29cb1f5ed13d6c29.jsonl`  
- SHA-256: `8dde02f8164a52ee4c047791eeab32e463b0a20fab43ba1c413423e47326809e`  
- Source modified: `2026-05-31T04:20:29+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:18:45.260Z)

Source Extractor

Research question: "Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON

transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/pdf/2508.05201>

****Title:**** FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance (ICAIF '25)

****Found via:**** Borderless-table accuracy & financial-number hallucination/verification (contrarian/skeptical) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-05-31T04:18:48.712Z)

I'll fetch the source and extract key claims relevant to the research question about local document-AI/OCR models for parsing bank statement PDFs.

3. user (2026-05-31T04:19:02.506Z)

Key Findings on Intrinsic Tabular Hallucinations in Finance

****Publication Details:****

- ****Title:**** FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance
- ****Authors:**** Mengao Zhang, Jiayu Fu, Tanya Warriier, Yuwen Wang, Tianhui Tan, Ke-wei Huang
- ****Venue:**** 6th ACM International Conference on AI in Finance (ICAIF '25), November 15-18, 2025, Singapore
- ****Date:**** October 27, 2025

(1) Problem Definition: Intrinsic Tabular Hallucinations

The research addresses how large language models and vision-language models produce erroneous

financial data when processing tables. The framework specifically focuses on understanding how models misrepresent numerical information in financial contexts without relying on external knowledge beyond training data.

(2) Types of Hallucinations on Financial Numbers

While the PDF metadata and structure indicate the paper examines how models hallucinate on financial numbers in tables, the specific truncated content prevents detailed extraction of explicit hallucination categories (transposed digits, dropped rows, duplicated rows, wrong values). The document references prior work on hallucination detection but the main findings section is in the compressed content streams.

(3) Quantitative Results on Hallucination Rates

The document structure indicates Tables 8, 9, 10, and 13 contain quantitative results, but the specific accuracy metrics and hallucination rates across models are contained in the compressed content streams that were truncated in the source material provided.

(4) Models Tested and Comparisons

The paper mentions testing multiple models, with references suggesting evaluation across different LLM and VLM architectures. However, the specific model names, versions, and comparative performance data are located in sections not fully readable from the truncated PDF metadata.

(5) FAITH Framework for Detection/Measurement

The framework title indicates a structured approach to "assessing intrinsic tabular hallucinations," suggesting a methodology for measurement. Keywords indicate focus on "benchmarking" and "model verification and validation," but detailed framework specifications are in inaccessible sections.

(6) Implications for Financial Table Verification

The research's positioning within finance applications and emphasis on hallucination assessment suggests implications for critical financial analysis, but specific recommendations are located in truncated sections.

****Note:**** The PDF content provided was heavily compressed with FlateDecode streams. A readable version would better support extraction of the quantitative results, specific models evaluated, and detailed framework specifications.

[Binary content (application/pdf, 1006.3KB) also saved to C:\Users\avidu\claude\projects\C--Users-avidu-Projects-muneem\9fd4426-f234-4e4a-bfb4-1f2c8255c661\tool-results\webfetch-1780201019681-c2bd5q.pdf]

4. assistant (2026-05-31T04:19:05.743Z)

The PDF content was compressed and the WebFetch couldn't extract the substantive findings. The PDF was saved locally. Let me read it directly to extract the actual claims. I'll also try the HTML abstract page in parallel for context.

5. user (2026-05-31T04:19:06.766Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. user (2026-05-31T04:19:17.607Z)

FAITH Framework Paper Summary

Abstract (Verbatim)

"Hallucination remains a critical challenge for deploying Large Language Models (LLMs) in finance. Accurate extraction and precise calculation from tabular data are essential for reliable financial analysis, since even minor numerical errors can undermine decision-making and regulatory compliance. Financial applications have unique requirements, often relying on context-dependent, numerical, and proprietary tabular data that existing hallucination benchmarks rarely capture. In this study, we develop a rigorous and scalable framework for evaluating intrinsic hallucinations in financial LLMs, conceptualized as a context-aware masked span prediction task over real-world financial documents. Our main contributions are: (1) a novel, automated dataset creation paradigm using a masking strategy; (2) a new hallucination evaluation dataset derived from S&P 500 annual reports; and (3) a comprehensive evaluation of intrinsic hallucination patterns in state-of-the-art LLMs on financial tabular data. Our work provides a robust methodology for in-house LLM evaluation and serves as a critical step toward building more trustworthy and reliable financial Generative AI systems."

Key Metadata

- **Venue:** AMC ICAIF'25
- **Submission Date:** August 7, 2025 (revised October 24, 2025)
- **Authors:** Mengao Zhang, Jiayu Fu, Tanya Warriar, Yuwen Wang, Tianhui Tan, Ke-wei Huang
- **Classification:** Machine Learning (cs.LG)

Notable Findings

The document describes the FAITH framework as implementing "a context-aware masked span prediction task" using real financial documents. However, the provided content contains no specific hallucination rates, detailed model comparisons, or empirical results?only the conceptual framework description.

7. assistant (2026-05-31T04:19:20.754Z)

The abstract gives me the core framing. The PDF needs a tool to extract text. Let me try Python (pdfplumber/pypdf are likely available given the project context) to extract the key results sections.

8. user (2026-05-31T04:19:22.239Z)

no pypdf No module named 'pypdf'
no pdfplumber No module named 'pdfplumber'

9. assistant (2026-05-31T04:19:24.622Z)

Not in the global env. The project uses these though. Let me find the project's Python environment.

10. user (2026-05-31T04:19:26.125Z)

```
Exit code 2
total 77
drwxr-xr-x 1 avidu 197609    0 May 31 08:52 .
drwxr-xr-x 1 avidu 197609    0 May 30 21:47 ..
drwxr-xr-x 1 avidu 197609    0 May 31 09:42 .git
-rw-r--r-- 1 avidu 197609 286 May 30 23:27 .gitattributes
drwxr-xr-x 1 avidu 197609    0 May 31 08:52 .github
drwxr-xr-x 1 avidu 197609    0 May 30 23:27 .github
-rw-r--r-- 1 avidu 197609 1403 May 31 08:39 .gitignore
```

```

drwxr-xr-x 1 auidu 197609      0 May 30 22:57 .pytest_cache
drwxr-xr-x 1 auidu 197609      0 May 30 22:57 .ruff_cache
drwxr-xr-x 1 auidu 197609      0 May 30 22:57 .venv
-rw-r--r-- 1 auidu 197609  4273 May 30 21:54 CLAUDE.md
-rw-r--r-- 1 auidu 197609 17507 May 30 23:43 DESIGN.md
-rw-r--r-- 1 auidu 197609  1080 May 31 08:52 Makefile
-rw-r--r-- 1 auidu 197609  2925 May 31 00:03 README.md
drwxr-xr-x 1 auidu 197609      0 May 31 09:42 docs
-rw-r--r-- 1 auidu 197609  1916 May 31 08:39 pyproject.toml
drwxr-xr-x 1 auidu 197609      0 May 30 22:57 src
drwxr-xr-x 1 auidu 197609      0 May 31 08:37 testdata
drwxr-xr-x 1 auidu 197609      0 May 31 08:52 tests
drwxr-xr-x 1 auidu 197609      0 May 31 08:39 tools
---VENV---
.venv/Scripts/python.exe
---POETRY---

```

11. assistant (2026-05-31T04:19:29.222Z)

There's a `.venv`. Let me use the project's Python to extract the PDF text.

12. user (2026-05-31T04:19:33.666Z)

PAGES 9

FAITH: A Framework for Assessing Intrinsic Tabular
Hallucinations in Finance

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NationalUniversityofSingapore NationalUniversityofSingapore NationalUniversityofSingapore
Singapore Singapore Singapore

Abstract AIinFinance(ICAIF'25), November15?18, 2025, Singapore, Singapore. ACM,

HallucinationremainsacriticalchallengefordeployingLargeLan-
NewYork, NY, USA, 9pages. <https://doi.org/10.1145/3768292.3770433>

guageModels(LLMs)infinance. Accurateextractionandprecisecal-
culationfromtabulardataareessentialforreliablefinancialanalysis, 1 Introduction

sinceevenminornumericalerrorsscanunderminedeconomic-making

WhileLargeLanguageModels(LLMs)arerapidlytransformingfi-

andregulatorycompliance. Financialapplicationshaveunique

nancialservices throughcapabilities such as automated information

quirements, often relying on context-dependent, numerical, and

extraction[4] and client-facing chatbots[16], their deployment also

proprietary tabular data that existing hallucination benchmarks

introduce substantial risks, among which hallucination poses seri-

rarely capture. In this study, we develop a rigorous and scalable

ousthreat to decision-making and stakeholder trust[8, 19?22]. In

framework for evaluating intrinsic hallucinations in financial LLMs,

response, regulatorssuch as the Monetary Authority of Singapore

conceptualized as a context-aware masked span prediction task over

(MAS) have underscored the need for robust model risk manage-

real-world financial documents. Our main contributions are: (1) a

ment framework tailored to advanced AI applications[15]. Despite

novel, automated dataset creation paradigm using a masking strat-

growing regulatory scrutiny, systematic methods for evaluating

egy; (2) a new hallucination evaluation dataset derived from S&P

and mitigating hallucinations in financial contexts remain largely

500 annual reports; 1 and (3) a comprehensive evaluation of intrinsic

undeveloped. As a foundational step, this study proposes a scal-

hallucination patterns in state-of-the-art LLMs on financial tabu-

able, comprehensive framework for evaluating and analyzing LLM

lar data. Our work provides a robust methodology for in-house hallucination in finance, quantifies how increasing information-LLM evaluation and serves as a critical step toward building more extraction complexity exacerbates hallucination errors, and offers trustworthy and reliable financial Generative AI systems. action-able guidance for researchers and practitioners. Recent empirical studies have begun to reveal that hallucinations CCSC concepts in finance exhibit domain-specific characteristics compared with ? Computing methodologies? Natural language processing; those observed in general-domain NLP benchmarks. For instance, Model verification and validation; ? Applied computing? Shah et al. [22] find that LLMs' factual accuracy on historical data Economics; ? Information systems? Information retrieval. varies across company size and financial periods, while Kang and Liu [8] and Sarmah et al. [21] demonstrate that even state-of-the-keywords art models frequently misinterpret financial terms or misreport Large Language Models, Hallucination, Benchmarking, Financial numerical values. Complementary works such as Roychowdhury NLP, Table Reasoning [19] outline stages toward halluci

13. assistant (2026-05-31T04:19:37.807Z)

Good, the text extraction works. Now let me extract the key results sections ? particularly the findings on how complexity increases hallucination, and any quantitative results.

14. user (2026-05-31T04:19:41.017Z)

nsichallucina- Yet, evaluating hallucinations in a systematic and robust manner patterns in state-of-the-art LLMs on financial tabular ner remains a non-trivial challenge due to the scarcity of finance-data, including a detailed breakdown by reasoning type. specific benchmarks. Existing datasets are predominantly derived from general-domain text such as Wikipedia [3, 6], whereas financial applications involve numerically grounded, context-dependent 2.1 Hallucination Evaluation Datasets data and specialized terminology. More importantly, there is a clear need for an automated and scalable approach to creating evaluation LLM hallucinations can be categorized as extrinsic or intrinsic, ation datasets for hallucination detection in real-world financial based on the reference source [3]. The evaluation of extrinsic hallucinations assesses whether a model's output, generated from keep up with the evolving LLM behaviors or the diversity of proprietary its internal knowledge alone, aligns with the established world facts. financial data. Benchmarks in this category [10, 11, 13] generally evaluate model To address these gaps, we develop FAITH (A Framework for Faithful AI) against static, open-domain corpora like Wikipedia. Financial Assessing Intrinsic Tabular Hallucinations in Finance) - an automated and scalable approach for constructing intrinsic hallucination time-sensitive, context-intensive, and often proprietary, requiring evaluation tasks directly from real financial annual reports, where evaluation methods that prioritize contextual accuracy over pre-ground-truth numerical values are explicitly available. We address a trained, parametric knowledge [8, 19?22]. core challenge in financial LLM applications: accurately identifying, Therefore, intrinsic hallucination evaluation is more relevant and calculating, and presenting numerical values based on financial critical for financial applications, as it directly measures a model's statements (input context). We introduce a novel context-aware faithfulness to a specific, provided context. While intrinsic evaluation masked span prediction task, where numerical values from real ation datasets have emerged, especially for Retrieval-Augmented

annual reports are masked and LLMs are prompted to recover them. Generation (RAG) applications [6, 14, 17], those based on general-To ensure reliable evaluation, we develop an novel filtering method purpose text, mostly are limited to simple look-up tasks rather than that select only those values with a unique, consistent, and answerable ground truth. A central contribution is our taxonomy of Our work addresses this critical gap, focusing on intrinsic hallucinations within complex financial tabular data: a unique and high-Direct Lookup, Comparative Calculation (The value is a function of stakes domain largely overlooked by current evaluation method-the same variable in different periods, e.g., year-over-year growth), Bivariate Calculation (the value is a function of two variables, e.g.,

2.2 Tabular Reasoning and Evaluation

gross margin), and Multivariate Calculation (more complex dependencies). This classification enables structured evaluation across Our focus on intrinsic hallucinations with tables connects to the reasoning complexity. Using our approach, we build a large-scale broader research area of tabular reasoning. Foundational benchmark dataset from S&P 500 annual reports and evaluate leading LLMs on their intrinsic hallucination rates. pivotal in assessing core LLM skills like numerical reasoning. In Empirically, our results reveal a stratified hierarchy of the more intricate financial domain, specialized benchmarks have model reliability in financial reasoning. Proprietary frontier emerged to address higher complexity. Datasets like FinQA [5] and models such as Claude-Sonnet-4 and Gemini-2.5-Pro achieve high TAT-QA [30] test joint reasoning over tables and text, while others overall accuracy, yet even these systems exhibit 10-20% error rates focus on navigating complex hierarchical structures [9, 23, 27]. on multi-step numerical reasoning. In contrast, smaller open-source While these benchmarks have significantly advanced LLM reasoning capabilities, their evaluation scope is primarily focused on near-zero in multivariate scenarios. This degradation with reason-the correctness of the final answer. This focus on accuracy creates ing complexity highlights an urgent limitation of current architectures: as task complexity increases, so does the risk of hallucination. Addressing this gap requires moving beyond traditional question-answering formats to develop scalable

===PAGE 3===

FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance

ICAF 2025, November 15-18, 2025, Singapore, Singapore

<LLM Answerability Annotation> <Format Input>

Text before: Services and other revenue consists of ?

Text before: Services and other revenue?

Sentence: Automotive sales revenue increased

\$11.30 billion, or 17%, Span1: Automotive sales revenue increased in the year ended December 31, 2023? SpanSplit <MASK> or 17% ?

Span2: ?increased \$11.30 billion or

Text after: The increase was partially offset by a lower

<MASK> in the year ended December 31?

average? For each highlighted span, determine

<Sentence Extraction and Numerical Span Identification> if the span is answerable? T b e y x a t a l o f t w e r e : r T a v h e e r i a n g c e r ? ease was partially offset

Span1 Span2 <Ground Truth>

Text Before: The following table disaggregates our Span1 Span2

revenue? \$11.30 billion 17%

10-K

Reports <LLM Fill in the Blank>

From SEC

Your task is to fill in the masked blank in a

sentence using the provided financial data tables. \$11.30 million 16.78%

<Context Extraction> <Prediction>

Figure 1: Illustration of the task definition and data processing.

methodologies that can directly probe for such intrinsic errors. Our $\mathcal{P}$, $\mathcal{M}$?

and replacing its content with a [MASK] token, producing

study proposes an novel framework to rigorously and automatically ing a corrupted sentence???.

evaluate intrinsic hallucination in financial tabular reasoning. The objective of the model?

is to recover the original content of

the masked span? $\mathcal{P}$, $\mathcal{M}$. The model is conditioned on the corrupted

2.3 Automatic Dataset Construction sentence and a context set? $\mathcal{P}$:

The high cost and limited scalability of manual annotation have

???, $\mathcal{P}$ = $\mathcal{P}$ (???, $\mathcal{P}$?) (1)

driven the development of automated evaluation dataset construc-

tion methods. Some approaches achieve fine-grained analysis by

decomposing generated content into atomic facts for verification

The context? $\mathcal{P}$ = {T, P, $\mathcal{P}$? $\mathcal{P}$?1, $\mathcal{P}$? $\mathcal{P}$?+1 } include the tables, their pre-

text, and the immediately preceding and succeeding sentences. The

[1, 7, 13], while others leverage the inherent structure of rela-

goals for the predicted span???, $\mathcal{P}$ to be matched with the original

tional data to automatically generate complex, verifiably question-

span?

$\mathcal{P}$, $\mathcal{P}$

answer pairs [18]. Adversarial test cases can be dynamically created

through data perturbation [26]. Although using LLMs as annota-

3.2 Masking Criteria

tors may introduce biases [28], recent work shows that carefully

designed automated pipelines can yield high-quality annotations,

To ensure the task setup yields meaningful and unbiased hallucina-

sometimes rivaling human performance [12]. We build on these

tive evaluation, three key assumptions must hold for each masked

advancements by adapting and extending automated benchmarking

span. These assumptions guide our masking strategy and directly

paradigm to the unique challenges of financial tabular analysis.

impact the reliability of evaluation outcomes:

? Uniqueness: The masked span must have a unique correct

3 Task Definition

answer, preventing multiple plausible completions.

In this section, we focus on the core requirement of retrieving in- ?

Example: A sentence like "The company's [MASK] has im-

formation from context and providing accurate answers in complex

proved "would be excluded, as several financial metrics such

financial scenarios, which is a common need for financial LLM

as "revenue," "profit margin," or "cash flow" could all be rea-

applications. We formulate the task as a context-aware masked so-
nable answers.

span prediction over tabular financial data, for the goal of intrinsic ? Consistency: The ground-

truth span must be consistent with

hallucination evaluation. Figure 1 demonstrates key flow.

the context, avoiding internal misalignments within the source

document.

3.1 Problem Formulation ? Example: If the table reports operating income as \$500 million,

Let a financial document? be composed of a set of structured tables

the masked sentences should not mistakenly state "Operating

T and a body of text. This text is partitioned into two distinct, non- income was \$500 thousand" as the to-be-

masked truth. Evalu-

ating based on the incorrect masked content would unfairly

overlapping sets:

penalize a model that correctly aligns with the table.

? A set of explanatory pre-texts $\mathcal{P}$, where each pre-text? $\mathcal{P}$? $\mathcal{P}$? $\mathcal{P}$?

Answerability: The masked span must be inferable from the

introduce the table? $\mathcal{P}$? T.

provided context, ensuring the task is solvable by an LLM.

? A sequence of general sentences? = ($\mathcal{P}$

1

, $\mathcal{P}$

2

, ..., ? ?), which con-
 ? Example: If the sentence states "The company's revenue in-
 stitute the remaining text of the document.
 created by [MASK] in 2024 compared to 2023, "both revenue
 Within a given sentence? ? ??, we identify a set of non-overlapping
 figures for 2024 and 2023 should be present in the table or
 spans of interest, $M = \{? ?, 1, ? ?, 2, \dots, ? ?, ?\}$, based on specific cri-
 teria. A task instance is constructed by selecting a single span
 answerable and should be excluded from evaluation.
 ===PAGE 4===
 ICAIF'25, November 15-18, 2025, Singapore, Singapore Zhang, et al.
 Algorithm 1 Precision-Relaxed Matching with Unit Groups
 (e.g., percentage or absolute value), thereby restoring uniqueness.
 Require: Ground truth span?, predicted span?, unit groups U To solve these second, we introduce a precision-
 relaxed evaluation
 Ensure: $??_????_????$ and $??_????_????$ protocol that normalizes and compares the numeric predictions.
 1: Let U scales ? U be the subset of scale units This protocol is detailed in Algorithm 1 and consists of two main
 2: ($??, ? ?$)? NormalizeNumber($?, U$ scales) components:
 3: ($??, ? ?$)? NormalizeNumber($??, U$ scales) Numeric Matching. This process first normalizes
 the ground-
 // Normalize to base value and determine precision
 truth span? and the predicted sp

15. assistant (2026-05-31T04:19:44.468Z)

This is rich. I now have the core empirical findings. Let me grab the results/discussion section (pages 5-8) where the quantitative breakdowns and key takeaways live, especially anything about VLMs and verification.

16. user (2026-05-31T04:19:49.653Z)

===PAGE 5===
 FAITH: A Framework for Assessing Intrinsic Tabular Hallucinations in Finance
 ICAIF'25, November 15-18, 2025, Singapore, Singapore
 Table 1: Agreement among LLMs and human annotation for Table 2: Comparative statistics of the Pilot
 (human-
 Answerability Annotation. "Yes" and "No" represent "answer- annotated) and Main (LLM-
 annotated) dataset splits.
 able" and "unanswerable" respectively. The LLMs used are
 GPT-4.1, Claude-sonnet-4, and Gemini-2.5-pro.
 Metric Pilot Split Main Split
 Number of companies 9 453
 Human
 Avg. context length (chars) 14,148 12,843
 LLM Agreement Pattern Yes No Total Avg. number of tables 14.9 19.2
 Number of sentences 164 1,122
 All 3 Agree (Unanimous)
 Number of answerable spans 300 2,406
 3 LLMs-No 1 626 627
 3 LLMs-Yes 276 11 287
 2-1 Split (Disagree)
 2 LLMs-Yes, 1 LLM-No 12 23 35 Results of Operations" (MD&A). This section is selected for its rich
 1 LLM-Yes, 2 LLMs-No 11 164 175 narrative analysis and manageable length, providing a dense source
 Total 300 824 1124 of contextualized financial data. We apply a keyword-based search
 to locate the MD&A section, supplementing this automated process
 with manual curation to ensure data quality.
 Discussion and Analysis" section of 10-K reports from nine compa-
 Context and Sentence Extraction. To avoid overlaps between
 nities across diverse industries (e.g., healthcare, finance, technology).
 context and sentence, we first extract tables and their immediate
 Each span is labeled for answerability by at least two financial
 preceding sentence as the explanatory pre-texts. Then we parse the
 experts, with a senior reviewer adjudicating disagreements. This
 rest of the content into plain text. After which a sentence split is

process yields a high inter-annotator agreement (Fleiss' Kappa = 0.905) and results in 300 answerable and 824 unanswerable spans. We incorporated custom rules to merge fragments that typically arose. The annotators also classify the 300 answerable spans into our four artifacts from the document conversion process. financial reasoning scenarios (A: Direct Lookup, B: Comparative, Numerical Span Identification. From this sentence corpus, C: Bivariate, D: Multivariate), resulting in a distribution of 212, 58, we isolate claims containing numerical data. This is a two-stage process: 22, and 8 respectively, which reflect the natural rarity of more complex reasoning tasks. Initial Detection: We employ spaCy's Named Entity Recognition (NER) to detect entities such as MONEY, PERCENT, CAR. Claude-Sonnet-4, Gemini-2.5-Pro) with the answerability annotation task. The results, summarized in Table 1, show that unanimous LLM consensus is an exceptionally strong indicator of answerability correctness. When all three models agreed, they included relevant currency symbols (e.g., \$) and a comprehensive vocabulary of financial units and scales (e.g., "million," "billion," and "per share"). This pilot study validates that leveraging unanimous LLM consensus is a highly reliable and scalable strategy for annotating LLM-based Answerability Annotation. To create a representative sample, we randomly select 10 sentences from each 10-K rather than generating new content; thus, using LLMs as an annotation tool does not introduce fabrications. This ensures the soundness of LLMs employed for annotating whether each span is answerable given the provided context. A span is retained in the final dataset only if all three models unanimously classified it as answerable.

4 Evaluation Dataset

4.2 Dataset Statistics

4.1 Data Collection and Processing

Table 2 presents a statistical overview of our dataset, which is composed of a human-annotated Pilot Split and a larger, LLM-annotated Main Split covering 453 S&P 500 companies after processing and financial reporting practices and covers a wide spectrum of industries. We source these documents in XBRL format from the SEC's EDGAR database. The collected filings then undergo several processing steps. A key characteristic of our dataset is its real

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ICAF 2025, November 15-18, 2025, Singapore, Singapore Zhang, et al.
Table 3: Model Accuracy (%) on Pilot and Main Splits, including scenario performance. Scenario abbreviations: (A) Direct Lookup, (B) Comparative Calculation, (C) Bivariate Calculation, and (D) Multivariate Calculation.

Model Pilot Split Main Split

Overall	Value	Unit	A	B	C	D	Overall	Value	Unit	A	B	C	D
(n=212)	(n=58)	(n=22)	(n=8)	(n=1606)	(n=635)	(n=135)	(n=10)						
Gemini-2.5-pro	95.0	95.7	97.7	96.7	91.4	95.5	75.0	91.9	97.8	93.1	91.8	94.0	96.3
Gemini-2.5-flash	91.0	92.3	96.7	94.8	82.8	90.9	50.0	88.7	91.1	93.7	90.2	88.0	87.4
Gemini-2.5-flash-lite	55.7	59.0	83.0	67.0	27.6	27.3	37.5	50.2	57.9	57.8	45.8	59.4	70.4
Claude-sonnet-4	93.0	93.7	97.3	96.2	93.1	81.8	37.5	95.6	95.8	98.7	97.0	82.6	80.0

Claude-haiku-3	64.7	66.0	92.7	71.7	50.0	59.1	0.0	81.3	81.8	93.9	84.1	80.6	66.7	30.0
GPT-4.1	89.7	90.3	95.3	93.9	87.9	72.7	37.5	89.2	89.5	92.6	91.1	90.4	77.8	30.0
GPT-4.1-mini	78.0	79.3	91.7	85.4	70.6	40.9	37.5	88.2	89.4	94.2	89.7	89.0	80.7	70.0
GPT-4.1-nano	31.3	43.3	62.3	36.8	19.0	22.7	0.0	70.0	71.9	92.1	71.8	72.3	53.3	10.0
Qwen-3-8B	20.3	32.0	44.3	24.5	10.3	13.6	0.0	30.6	35.6	36.1	27.1	35.7	54.1	0.0
Qwen-3-32B	68.0	70.0	96.0	80.7	34.5	45.5	37.5	73.9	76.7	83.7	73.5	76.1	81.5	40.0
Ministral-8B	22.3	23.0	75.7	29.7	1.7	9.1	12.5	40.8	41.8	74.7	45.3	33.2	31.9	0.0
Mistral-small-24B	63.7	63.7	97.7	77.8	29.3	31.8	25.0	45.6	64.2	53.4	45.3	46.0	57.0	0.0
Llama-3.1-8B	27.0	29.0	68.3	35.3	6.9	9.1	0.0	47.5	47.8	85.5	47.8	49.9	43.7	10.0
Llama-3.3-70B	56.0	57.0	95.3	66.0	36.2	31.8	0.0	37.0	40.9	42.5	34.7	40.9	53.3	10.0
Gemma-3-12B	32.0	34.3	83.7	42.5	6.9	9.1	0.0	15.2	19.7	37.7	16.3	12.8	17.0	0.0
Gemma-3-27B	37.3	39.0	91.0	45.7	20.7	13.6	0.0	33.8	37.3	51.0	31.7	38.6	43.7	0.0

5 Experiments 5.2.1 AQuantifiableHierarchyofModelReliability. Therresults demonstrateadistinctperformancehierarchyamongtheevaluated

5.1 ExperimentalSetup models.

Weevaluatearangeofstate-of-the-artopen-sourceandproprietary Atoptierofproprietarymodels, ledbyClaude-Sonnet-4(95.6% LLMsonourdataset?spilotandmainsplits.Duringevaluation, onMainSplit)andGemini-2.5-Pro(91.9%onMainSplit), exhibits wepromptthemodelstogenerateastep-by-steprationalebefore highoverallaccuracy.Theirstrongperformance,whichremains predictingthefinalmaskedvalueandtoself-classifytheirreasoning largelyconsistentbetweenthePilotandMainsplits, validatesthe processintooneofthefourfinancialscenarios(A-D).Thedetailed robustnessofourbenchmark.However, the4-8%errorrate, while promptisavailableinAppendixA.1. lowrelativetoothermodels, remainsasignificantconsideration Accuracyisusedastheprimaryevaluationmetric, computed forfinancialapplicationswhereprecisionisnon-negotiable. foroverallpredictionsandseparatelyforthenumericvalueand Belowthistoptier, weobserveasecondgroupofcapablemod- unitcomponents.Inaddition, wereporttheaccuracystratifiedby els, includingGPT-4.1(89.2%)andGPT-4.1-mini(88.2%), which financialscenario.Forthepilotsplit, scenariolabelsareassigned deliverrespectablebutdemonstrablyloweraccuracy.Afurther byhumanannotators.Forthemainsplit, weusethestrategystated sharpdeclineisevidentinthemajorityofothermodels, particu- insection3.4.Thisstratificationallowsforadetailedanalysisof larlysmalleropen-sourcevariants.ModelssuchasLlama-3.1-8B errortypes. (47.5%)andQwen-3-8B(30.6%)exhibithigherrorfrequencies, ren- Allmodelsarerunwithatemperatureof0toensurereproducibil- deringthemunsuitablefortasksrequiringfinancialfidelity. ity.ProprietarymodelsareaccessedviatheirofficialAPIs, while open-sourcemodelsarerunon4H200GPUsusingtheLLaMA- Factoryframework[29].Weimplementaretrymechanism(upto 5.2.2 Reasoning Complexity as the Primary Performance Differ-

3attempts)tohandleinstanceswhereamodelfailedtoproducea entiator. Theanalysisofperformanceacrossthefourreasoning validJSONoutput. scenarios(A-D)revealsthattaskcomplexityisthemostsignificant factorinfluencingmodelreliability.

DegradationinMulti-stepScenarios. Whilemostmodelsperform 5.2 ExperimentResults

adequately on Direct Lookup (A), accuracy systematically de- This section presents a detailed analysis of the performance of creasesastasksrequirecalculationandlogicalinference.Themost variousLLMsonourfinancialhallucinationbenchmark.Thecom- pronouncedperformancedropocc

===PAGE 7 ===

FAITH:AFrameworkforAssessingIntrinsicTabularHallucinationsinFinance ICAIF?25, November15?18, 2025, Singapore, Singapore Table4:KeyfinancialattributesforMohawkCommonsused Incontrast, GPT-4.1andClaude- Sonnet-4failedtoutilizethe inthemaskevalueinferencetask.(Tableheadersandrows

debt information from the table. Instead, both models based their predictions solely on the information in the sentence, by simply calculating 90% of the purchase price ($62.1M \times 0.90 = 55.9M$). This neglects the mortgage debt and indicates insufficient integration of tabular data.

Gotham 49% \$8.5 8.36% Mar 2024

Implications for Multimodal Reasoning: This example re-

Mohawk Commons 18.1% \$7.2 5.80% Mar 2028

veals a key limitation in current LLM capabilities: the inference of total \$188.8 latent variables that requires synthesizing information across modalities and reasoning over implicit relationships. Only Gemini 2.5 Pro reconstructed the correct financial logic chain, whereas other models defaulted to predicting the masked span without grounding resilience in top-tier models. In contrast, the top-performing answers in the provided numerical data.

models demonstrate notable, albeit imperfect, resilience on complex tasks. Gemini-2.5-Pro is particularly robust in the Multivariate Calculation (D) scenario, achieving 90.0% accuracy on the Main

6 Conclusion and Future Work

Split. Claude-Sonnet-4 also performs strongly at 80.0%. While these results are promising and highlight advanced reasoning capabilities, a 10-20% failure rate on the most complex calculations work for evaluating hallucinations of LLMs in the financial domain, represents a critical barrier. This difficulty in maintaining factual providing an nuanced view of their current capabilities. Our findings consistency through multi-step logic remains a primary vector for indicating that even state-of-the-art models struggle with the nuances

intrinsic hallucinations. off financial tabular data. While leading models are approaching the accuracy required for less critical applications, the fundamen-

5.2.3 Case Study. Beyond aggregate metrics, we perform a qualitative challenges of ensuring factual integrity, particularly regarding tative analysis to diagnose the models' specific failure modes. Our numerical scale in complex reasoning, remain the primary barrier analysis reveals several recurring patterns, with one of the most to their safe deployment in accuracy-critical financial workflows. significant being scale error. This occurs when a model correctly

For future work, we plan to expand the benchmarking dataset to identifies a numerical value but fails to associate it with the proper include more document types and more complex reasoning scenarios. magnitude (e.g., reporting "\$150" instead of "\$150 million"). Correct- ios. Another promising direction is to study how factors like table ing for this single error type in llama-3.3-70B's outputs, for instance, size and the number of tables in the context affect hallucination would boost its value accuracy from 37.0% to 57.7%, highlighting rates. In summary, we believe that robust, domain-specific evaluation is a critical vulnerability in contextual numerical grounding. tion is a crucial step towards building more reliable and trustworthy However, more profound failures stem from a deficient under- LLMs for financial applications.

standing of financial concepts that require multi-step reasoning across different data sources. To illustrate these deeper challenges, we now present a representative detailed case study on a sample

7 Limitation

that required latent variable inference from both tabular and textual context. The target span was the equity investment value (\$20.2 million) in the following scenario: tabular hallucinations in financial contexts, yet several limitations remain.

On January 27, 2023, FundVacquired a 90% interest Our benchmark draws on public U.S. company filings and curated financial tables. This design ensures data reliability and com-purchased a shopping center, Mohawk Commons, lo-

parability but limits coverage of diverse reporting styles, regional cated in Schenectady, New York, for \$62.1 million, in-accounting standards, and sector-specific terminologies. Nevertheless, the framework is readily expandable to other companies, Table 4 contains the relevant details for Mohawk Commons, industries, and jurisdictions, provided that structured financial data in

17. assistant (2026-05-31T04:20:01.521Z)

I now have the full picture of the empirical findings. I have everything I need: the quantitative model table, the "scale error" finding (highly relevant to muneem's transposed-magnitude concern), the degradation-with-complexity finding, and the specific open-source model failures. Let me extract the final claims.

Key findings relevant to the muneem research question:

1. **Scale errors** (correct digits, wrong magnitude ? e.g. "\$150" vs "\$150 million") are a named, recurring failure mode. Correcting this one error type lifted Llama-3.3-70B value accuracy from 37.0% to 57.7%. This directly bears on muneem's concern about transposed/wrong-magnitude numbers and validates magnitude-checking in verification.
2. **Even frontier models have 10-20% error on multi-step numerical reasoning**, and 4-8% overall error ? too high for "accuracy-critical financial workflows" without verification.
3. **Small open-source models fail catastrophically** on complex calculations (Qwen-3-8B 30.6%, Llama-3.1-8B 47.5%, Gemma-3-12B 15.2% overall on Main Split; near-zero on multivariate). This is highly relevant to the local-model recommendation ? small local LLMs are unreliable for financial number extraction/reasoning.
4. **Hallucination scales with task complexity** ? Direct Lookup is much more reliable than calculation. For muneem, pure extraction (lookup) is safer than asking a model to compute.
5. **Models defaulted to ungrounded predictions** rather than reading the table ? a hallucination pattern directly relevant to VLM-on-page-image risk.

Note one caveat for accuracy: this paper tests *text-based LLMs* reading provided text/table context* (masked span prediction), not vision/OCR of page images. The "intrinsic hallucination on financial tabular reasoning" finding generalizes, but it is not an OCR/VLM-vision benchmark. I'll flag this in the claims framing.

18. user (2026-05-31T04:20:17.977Z)

Structured output provided successfully

19. assistant (2026-05-31T04:20:29.848Z)

I have completed the analysis. Summary of what I found and how I handled it:

Source: FAITH (ICAIF '25, National University of Singapore ? Asian Institute of Digital Finance). Peer-reviewed primary research, published in an ACM conference. High quality.

Fetch note: The initial WebFetch could not read the PDF (FlateDecode-compressed streams), returning only the abstract-level framing. I extracted the actual results by reading the locally-saved PDF with the project's `.venv`pdfplumber` (global Python lacked it). This surfaced the quantitative model table and the discussion/case-study sections that contain the load-bearing claims.

Relevance caveat (important for synthesis): This paper benchmarks *text-based LLMs* on a masked-span-prediction task where the table/text context is *provided as text* ? it is **not** an OCR or vision-VLM-on-page-image benchmark. So it does not directly measure the OCR/layout error surface muneem cares about. What it does measure rigorously is *intrinsic numerical hallucination on financial tabular reasoning*, which generalizes to any model muneem feeds

statement data to. The findings most useful to the research question:

1. Frontier models still hit 10-20% error on multi-step numerical reasoning ? never trust raw output; muneem's balance-reconciliation verification is the right posture.
2. Small open-source models (8B-24B: Qwen-3-8B 30.6%, Llama-3.1-8B 47.5%, Gemma-3-12B 15.2%) fail catastrophically ? a concrete caution against small/quantized local LLMs for financial-number work.
3. "Scale error" (right digits, wrong magnitude) is a top failure mode ? muneem should verify units/magnitude, not just digit sequences.
4. Error scales with reasoning complexity ? use the model as a pure extractor, not a calculator.
5. Models default to ungrounded predictions instead of reading the table ? the core hallucination risk for any page-image?JSON approach.

These are returned in the structured output. No files were written.